Hints: Polynomials – 6. Polynomials over finite fields

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Count the common zeros modulo p: $N \equiv \sum_{x \in \mathbb{F}_p^n} \prod_{i=1}^{r} (1 - f_i(x)^{p-1}) \pmod p$, since each summand is 1 at a common zero and 0 otherwise.

Hint 2. The product has degree $\leqslant (p-1)\sum \deg f_i < (p-1)n$, so every monomial $x_1^{e_1} \cdots x_n^{e_n}$ in it has some $e_j < p-1$. Since $\sum_{t \in \mathbb{F}_p} t^e = 0$ for $0 \leqslant e < p-1$ (with $0^0 = 1$), every monomial sums to 0. Hence $p \mid N$, and $N \geqslant 1$ gives $N \geqslant p$.

Hint 3. EGZ for $2p-1$ integers $a_1, \ldots, a_{2p-1}$: apply the theorem to $\sum a_i x_i^{p-1}$ and $\sum x_i^{p-1}$ in $2p-1$ variables (degree sum $2p-2$, common zero 0). A nonzero common zero has $|\{i : x_i \neq 0\}| \equiv 0 \pmod p$, so exactly p nonzero coordinates, and the corresponding a_i sum to 0 mod p.

Problem 2

Hint 1. Let $|K| = q$ and work in the field $L = K[X]/(f) \cong \mathbb{F}_{q^n}$ with α the class of X. You must show $g(\alpha) = 1$.

Hint 2. The map $h \mapsto h(\alpha)$ is a bijection from polynomials of degree $\leqslant n-1$ onto L, and it sends the constants onto K. So $g(\alpha) = \prod_{\beta \in L \setminus K} \beta = \prod_{L^*} \beta \,/\, \prod_{K^*} \beta$.

Hint 3. By Wilson's theorem in a finite field, the product of all nonzero elements is -1 in both L and K.

Problem 3

(a)

Hint 1. Put $Y = X^q - X$. Then $Y^q - Y = \prod_{c \in K}(Y - c) = \prod_{c \in K}(X^q - X - c)$, and f_a is the factor with $c = -a$.

(b)

Hint 1. $f_a' = -1$, so f_a has q distinct roots in its splitting field; if θ is one, the roots are $\theta + c$, $c \in K$.

Hint 2. The degree of the irreducible factor through θ is the size of the orbit of θ under $\sigma(u) = u^q$. Here $\sigma(\theta) = \theta^q = \theta - a$, so $\sigma^k(\theta) = \theta - ka$.

Hint 3. For $a \neq 0$ the orbit has exactly p elements, so f_a is a product of $q/p = p^{n-1}$ distinct irreducible factors of degree p; for $a = 0$ it splits into q linear factors.

Problem 4

Hint 1. If $1 + 1 = 0$, the Frobenius $u \mapsto u^2$ is a bijection of K, so every coefficient is a square: $f(X^2) = \left(\sum b_i X^i\right)^2$ with $b_i^2 = a_i$.

Hint 2. Conversely, for odd $q = |K|$ you need some f of degree > 1 with $f(X^2)$ irreducible. If f is irreducible with root β and β is not a square in $K(\beta)$, then $X^2 - \beta$ is irreducible over $K(\beta)$, so a root of $f(X^2)$ has degree $2 \deg f$ over K.

Hint 3. Take β a generator of $\mathbb{F}_{q^2}^*$: it is not in K, so its minimal polynomial f has degree 2, and it is not a square because q^2-1 is even.

Problem 5

Hint 1. There are $q-1$ coefficients, nonzero and pairwise distinct, so they are exactly the elements of K^* in some order. Is 0 a root? Which $q-2$ roots can occur?

Hint 2. 0 is not a root, so the roots are $K^*\backslash\{s\}$ and $P = a\cdot\frac{X^{q-1}-1}{X-s} = a\,(X^{q-2}+sX^{q-3}+\cdots+s^{q-2})$, using $s^{q-1}=1$.

Hint 3. The coefficients $a, as, \ldots, as^{q-2}$ are pairwise distinct iff s generates K^*. Answer: $(q-1)\,\varphi(q-1)$.

Problem 6

Hint 1. Let $\phi(x) = x^p$, an $\mathbb{F}_p$-linear map of K. Then $f_m(x) = (\phi-1)^m(x)$, so $f_m(x) = f_1(f_{m-1}(x))$ and the roots of f_m form the kernel V_m of $(\phi-1)^m$, an $\mathbb{F}_p$-subspace. Roots of $f_1 = X^p - X$: exactly $\mathbb{F}_p$.

Hint 2. In characteristic p, $(\phi-1)^{p^j} = \phi^{p^j} - 1$, so $f_{p^j} = X^{p^{p^j}} - X$ and its roots in K form the subfield $\mathbb{F}_{p^{\gcd(n,p^j)}}$. Let p^s be the largest power of p dividing n and $L = \mathbb{F}_{p^{p^s}} \subseteq K$. Then every $V_m \subseteq L$, and $V_m = L$ for $m \geqslant p^s$.

Hint 3. On L, $N = \phi - 1$ satisfies $N^{p^s} = 0$ and $\dim\ker N = 1$; a nilpotent map with one-dimensional kernel is a single Jordan block, so $\dim V_m = \min(m, p^s)$. Answer: the roots of f_m form the $\mathbb{F}_p$-subspace $V_m = \{x : f_{m-1}(x) \in \mathbb{F}_p\}$ of L with $p^{\min(m,p^s)}$ elements, which is all of L once $m \geqslant p^s$.

Problem 7

Hint 1. Over $\mathbb{F}_2$ a polynomial of degree 2 or 3 is irreducible iff it has no root, i.e. constant term 1 and an odd number of terms. This gives x^2+x+1, x^3+x+1, x^3+x^2+1.

Hint 2. For degree 6 use $\frac{1}{n}\sum_{d|n}\mu(d)2^{n/d}$: $\frac{1}{6}(2^6-2^3-2^2+2) = 9$.

Problem 8

Hint 1. Run the Euclidean algorithm in $\mathbb{F}_7[x]$ (normalize leading coefficients using inverses mod 7).

Hint 2. The gcd is x^2+2: indeed $4x^4-2x^2+1 = (x^2+2)(4x^2+4)$ and $-3x^3+4x^2+x+1 = (x^2+2)(-3x+4)$ in $\mathbb{F}_7[x]$, and the cofactors are coprime.

Problem 9

Hint 1. The field must have at least 2022 elements; the smallest prime power $\geqslant 2022$ is 2027 (a prime). Can it be achieved, given that in characteristic 0 only $c(x-a)^n$ is divisible by its derivative?

Hint 2. In characteristic p, p-th powers have zero derivative. For $U \in \mathbb{F}_p[X]$, $(U^p)' = 0$.

Hint 3. With $p = 2027$ take $f = (X-a)\,U(X)^p$ where U has 2021 distinct roots in $\mathbb{F}_p$ other than a: then $f' = U^p$ divides f, and f has exactly 2022 roots. Answer: 2027.

Problem 10

Hint 1. Use the Taylor expansion $h(x+t) = h(x) + t\,h'(x) + t^2(\ldots)$ with an integer-coefficient remainder (the numbers $h^{(j)}/j!$ have integer coefficients).

Hint 2. From the hypothesis: h is a permutation mod p (values mod p depend only on x mod p, and each residue mod p must be hit), and $p \nmid h'(x)$ for all x, since otherwise $h(x+p) \equiv h(x) + ph'(x) \equiv h(x) \pmod{p^2}$.

Hint 3. If $h(x) \equiv h(y) \pmod{p^3}$ with $0 \leqslant x, y < p^3$, then $x \equiv y \pmod{p^2}$ (as $h \bmod p^2$ has period p^2). Write $y = x + kp^2$: $h(y) \equiv h(x) + kp^2 h'(x) \pmod{p^3}$, so $p \mid k$.